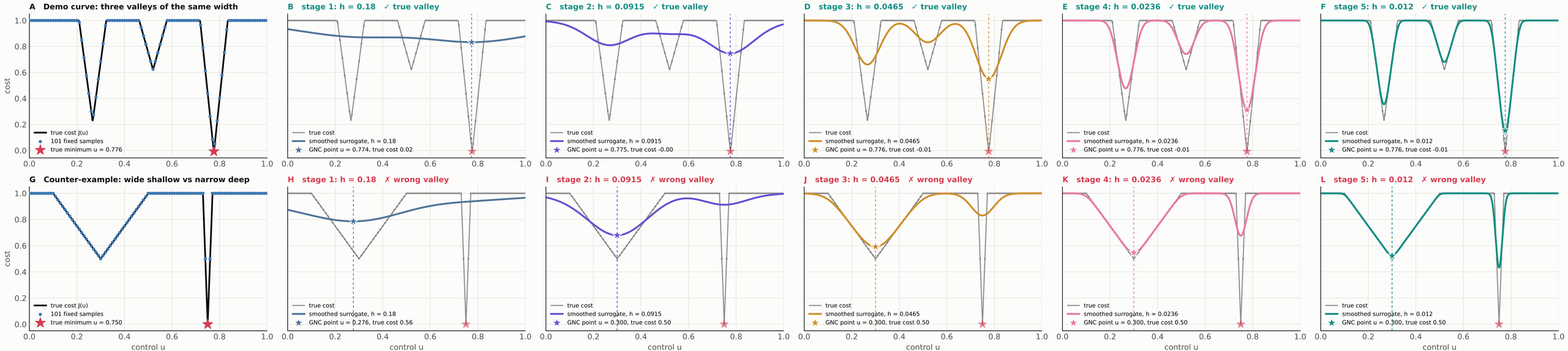

Kernel-surrogate GNC, 5 stages: same 101 samples at every stage, bandwidth 0.18 → 0.0915 → 0.0465 → 0.0236 → 0.012
Top: the demo's curve; smoothing finds the deepest valley because all valleys have the same width. Bottom: the smoothed average scores a valley by its area, so the wide shallow valley wins and the continuation stays there. Red star = true minimum.
A Demo curve: three valleys of the same width
B stage 1: h = 0.18 ✓ true valley
C stage 2: h = 0.0915 ✓ true valley
D stage 3: h = 0.0465 ✓ true valley
E stage 4: h = 0.0236 ✓ true valley
F stage 5: h = 0.012 ✓ true valley
G Counter-example: wide shallow vs narrow deep
H stage 1: h = 0.18 ✗ wrong valley
I stage 2: h = 0.0915 ✗ wrong valley
J stage 3: h = 0.0465 ✗ wrong valley
K stage 4: h = 0.0236 ✗ wrong valley
L stage 5: h = 0.012 ✓ true valley
cost
control u
true cost J(u)
101 fixed samples
true minimum u = 0.776
true minimum u = 0.750
true cost
smoothed surrogate, h = 0.18
GNC point u = 0.774, true cost 0.02
smoothed surrogate, h = 0.0915
GNC point u = 0.775, true cost -0.00
smoothed surrogate, h = 0.0465
GNC point u = 0.776, true cost -0.01
smoothed surrogate, h = 0.0236
GNC point u = 0.776, true cost -0.01
smoothed surrogate, h = 0.012
GNC point u = 0.776, true cost -0.01
GNC point u = 0.276, true cost 0.56
GNC point u = 0.300, true cost 0.50